

Qatar-1b

R-2829

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130716 · created 2026-09-05T23:27:38+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456489.707586 +/- 8e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.056
Transit depth (Rp/Rs)^2	3.61 +/- 2.15 [%]
Orbital Inclination (inc)	82.78 +/- 2.85
Airmass coefficient 1 (a1)	0.089 +/- 0.048
Airmass coefficient 2 (a2)	1.68 +/- 0.78
Scatter in the residuals of the lightcurve fit is	50.52 %
Best Comparison Star	#2 - [551, 426]
Optimal Aperture	1.46
Optimal Annulus	7.25
Transit Duration (day)	0.058 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.056 vs 0.1463 (deviation 0.78σ)
Duration fit vs published	0.058 d vs 0.0692 d (deviation 0.47σ)
Mid-time O–C	0.1 min
Depth SNR	3.4
Residual scatter	50.52 %

Light curve

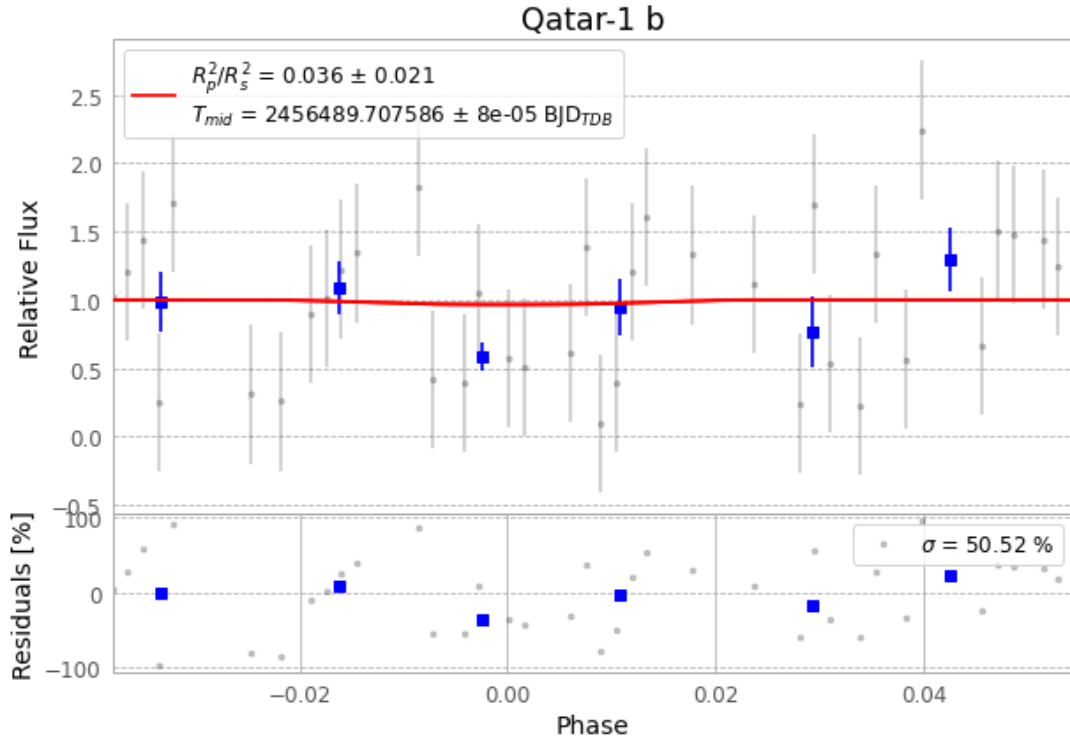


Figure 1 — FinalLightCurve_Qatar-1 b_2013-07-15.png (SHA-256 7eaf28396c8ea4db...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [565, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fbd9d45ee9197572...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), Qatar-1130716033912.FITS → Qatar-1130716065413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130716.log (273604ec46e7...), FinalLightCurve_Qatar-1 b_2013-07-15.png (7eaf28396c8e...), FinalLightCurve_Qatar-1 b_2013-07-15.csv (b58c079c65f2...)

Compute resources

Wall time 151.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 23:27Z.